

SAFETY DATA SHEET

SpecBio® Ectoine

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier:

Trade Name: SpecBio® Ectoine

Substance Name: (S)-2-methyl-3,4,5,6-tetrahydropyrimidine-4-carboxylic acid

CAS No.: 96702-03-3

EC No.: 431-910-1

Index No.: --

REACH Registration No.: *This substance is exempted from Registration as < 1t/a.*

1.2 Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Raw material for Cosmetic

Uses advised against: No data available

1.3 Details of the supplier of the safety data sheet

Manufactured By: Spec-Chem Industry Inc.
No.10 Wanshou Road (ShiLin Industrial Park)
Nanjing 211800, P.R. of China

Phone Number: 86-25-84523390, 84523391

Fax Number: 86-25-84520790, 84520791

Email: sc@specchemind.com

1.4 Emergency telephone number:

+49 1704724533 (European Union)

1-800-424-9300/ +1 703-527-3887 (North America, CCN 1010066)

SECTION 2: HAZARDOUS IDENTIFICATION

2.1 Classification of the substance or mixture

2.1.1 Classification according to Regulation (EC) No 1272/2008 (CLP)

Not a hazardous substance or mixture according to Regulation (EC) No. 1272/2008.

2.1.2 Additional information:

No more data.

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 [CLP] The product does not need to be labelled in accordance with EC directives or respective national law.

2.3 Other hazards

No data available

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Substance

Substance name	INCI Name	CAS No.	EC No.	Index number in CLP Annex VI	Weight % content
(S)-2-methyl-3,4,5,6-tetrahydropyrimidine-4-carboxylic acid	Ectoin	96702-03-3	431-910-1	--	≤ 100

SECTION 4: FIRST AID MEASURES

4.1 Description of first aid measures

Consult a physician. Show this safety data sheet to the doctor in attendance. Move out of dangerous area.

If inhaled

If symptoms develop, move individual away from exposure and into fresh air. If symptoms persist, seek medical attention. If breathing is difficult, administer oxygen. Keep person warm and quiet; seek immediate medical attention.

In case of skin contact

Wash off with soap and plenty of water. Take off contaminated clothing and wash it before reuse.

In case of eye contact

Flush eyes gently with water for at least 15 minutes while holding eyelids apart: seek immediate medical attention.

If swallowed

Never give anything by mouth to an unconscious person. Rinse mouth with water.

4.2 Most important symptoms and effects, both acute and delayed

No data available.

4.3 Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

SECTION 5: FIRE FIGHTING MEASURES

5.1 Extinguishing Media:

Suitable extinguishing media: Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

Unsuitable extinguishing media: high volume water jet.

5.2 Special Hazards arising from the substance or mixture

No data available.

5.3 Advice for Fire Fighters

Wear protective clothing to prevent contact with skin and eyes.

Wear self-contained breathing apparatus.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

6.1.1 For non-emergency personnel

Protective equipment: Wear the recommended personal protective equipment. For more information, see section 8: "Exposure controls / personal protection".

Emergency procedures: Ventilate the spill area. Keep unnecessary staff away.

6.1.2 For emergency responders:

Protective equipment: Do not handle without suitable protective equipment. Provide adequate protection for cleaner. For more information, see section 8: "Exposure controls / personal protection".

Emergency procedures: Ensure well ventilated.

6.2 Environmental precautions

Prevent further leakage or spillage if safe to do so. Do not empty into drains/surface water/ground water.

6.3 Methods and materials for containment and cleaning up

Collect and arrange disposal. Keep the chemical in suitable and closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Adhered or collected material should be promptly disposed of, in accordance with appropriate laws and regulations.

6.4 Reference to other sections

For disposal see section 13.

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for safe handling

Protective measures: Do not handle or use product until safety precautions recommended in the SDS have been read and fully understood. Wear personal protective equipment. Ensure good ventilation to avoid the formation of dust. Avoid open flames.

Advice on general occupational hygiene: Wash hands and other exposed areas thoroughly after handling. Do not eat, drink or smoke while using this product.

7.2 Conditions for safe storage, including any incompatibilities

Technical measures and storage conditions: Store in cool place. Keep container tightly closed in a dry and well-ventilated place.

Packaging materials: Inner: Aluminum foil bag/plastic bag; Outer: carton box

7.3 Specific end uses

no data available

SECTION 8: EXPOSURE CONTROLS / PERSONAL PROTECTION

8.1 Control Parameters

8.1.1. National occupational / biological exposure limit values

No data available

8.1.2. Information on monitoring procedures

No data available

8.1.3. Air contaminants formed

No data available

8.1.4. DNELs / PNECs values

No data available

8.1.5. Control banding

No data available

8.2 Personal protective equipment

8.2.1. Appropriate engineering controls

Ensure the workplace is well ventilated.

8.2.2. Individual protection measures, such as personal protective equipment

Avoid unnecessary exposure.

8.2.2.1 Eye and face protection: Safety glasses with side shields. Wear eye protection according to EN 166.

8.2.2.2 Skin protection:

Hand protection: Protective gloves made of plastic or rubber. Wear gloves that have been tested according to the EN 374 standard.

Other skin protection: Wear suitable protective clothing when working

8.2.2.3 Respiratory protection: Respirator with organic vapor cartridge. Not applicable with adequate ventilation.

8.2.2.4 Thermal hazards: No further information available

8.2.3. Environmental exposure controls

Environmental exposure controls: Avoid release to the environment.

Other Information: Do not eat, drink or smoke while using.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Physical state	Crystal or powder
Colour	White or almost white
Odour	No data available
Melting point	No data available
Boiling point	No data available
Flammability	No data available
Lower and upper explosion limit	Not applicable
Flash point	Not applicable
Auto-ignition temperature	No data available
Decomposition temperature	No data available

pH (2%)	5.0 - 8.0
Viscosity	Not applicable
Solubility	Water solubility: 390 g/L (23 °C)
Partition coefficient n-octanol/water	Log Pow: ≥ -3
Specific rotation	+139° - +145°
Vapour pressure	< 0 Pa (25 °C)
Density	1.568 g/cm ³ (20 °C)
Relative vapour density	Not applicable
Particle size	No data available
Surface tension	69.4 mN/m (20 °C, 0.1 vol%)

9.2 Other Information

9.2.1. Information with regard to physical hazard classes

No data available.

9.2.2. Other safety characteristics

No additional information available.

SECTION 10: STABILITY AND REACTIVITY

10.1 Reactivity

No unusual reactivity

10.2 Chemical stability

Stable under recommended storage conditions.

10.3 Possibility of hazardous reactions

No data available

10.4 Conditions to avoid

Strong heating.

10.5 Incompatibilities materials

Strong oxidizing agents.

10.6 Hazardous decomposition products

No information available.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Oral: OECD 401, Wistar rat, oral: gavage, LD50: > 2 000 mg/kg/dw.

Inhalation: No data available.

Dermal: OECD 402, Wistar rat, occlusive, LD50: > 2 000 mg/kg/dw.

Skin corrosion/irritation

In vivo skin irritation: OECD 404, New Zealand White rabbit, semioclusive, not irritating.

Serious eye damage/eye irritation

In vivo eye irritation: OECD 405, New Zealand White rabbit, not irritating.

Respiratory or skin sensitization

In vivo skin sensitisation: OECD 406, guinea pig, not sensitizing.

Germ cell mutagenicity

In vitro gene mutation study in bacteria: OECD 471, *S. typhimurium* TA 1535, .TA 1537, TA 1538, TA 98, TA100 and *E. coli* WP2, negative

In Vitro Mammalian Cell Gene Mutation Test: OECD 476, mouse lymphoma L5178Y cells, negative.

Carcinogenicity

No data available.

Reproductive toxicity

No data available.

Specific target organ toxicity - single exposure (Globally Harmonized System)

No significant toxic effects were observed in above acute oral and dermal tests.

Specific target organ toxicity - repeated exposure (Globally Harmonized System)

Short-term repeated dose oral toxicity (28 days): OECD 407, rat, oral: gavage, NOAEL: 300 mg/kg bw/day (actual dose received), NOEL: 300 mg/kg bw/day (actual dose received).

Aspiration hazard

No data available

SECTION 12: ECOLOGICAL INFORMATION**12.1 Toxicity****Acute toxicity:**

Fish: OECD 203, *Danio rerio*, semi-static, freshwater, 96h-NOEC: > 100 mg/L.

Aquatic invertebrates: OECD 202, *Daphnia magna*, static, freshwater, 48h-EC50: 100 mg/L.

Aquatic algae and cyanobacteria: OECD 201, *Desmodesmus subspicatus*, static, freshwater, 72h-EC50 (biomass/growth rate): > 100 mg/L, 72h-NOEC: > 100 mg/L.

Long-term toxicity:

No data available.

12.2 Persistence and degradability

Ready biodegradability: EU Method C.4-E, the degradation is 93% at 28 day, ready biodegradable.

12.3 Bioaccumulative potential

No data available

12.4 Mobility in soil

Adsorption / desorption: Koc: 2.25

12.5 PBT and vPvB assessment

Not a PBT or vPvB substance.

12.6 Endocrine disrupting properties

No data available

12.7 Other adverse effects

Toxicity to microorganisms: OECD 209, activated sludge, domestic, static, freshwater, 3h-EC50: > 1 000 mg/L.

SECTION 13: DISPOSAL CONSIDERATION

13.1 Waste treatment methods

Product

Dispose in accordance with local/ regional/national/international regulations. Offer surplus and non-recyclable solutions to a licensed disposal company. Dissolve or mix the material with a combustible solvent and burn in a chemical incinerator equipped with an afterburner and scrubber.

Contaminated packaging

Dispose in accordance with local/ regional/national/international regulations. Containers can be triply rinsed (or equivalent) and offered for recycling or reconditioning. Alternatively, the packaging can be punctured to make it unusable for other purposes and then be disposed of in a sanitary landfill. Controlled incineration with flue gas scrubbing is possible for combustible packaging materials.

SECTION 14: TRANSPROT INFORMATION

14.1 UN number:

ADR/RID: - IMDG: - IATA: -

14.2 UN proper shipping name

ADR/RID: Not dangerous goods

IMDG: Not dangerous goods

IATA: Not dangerous goods

14.3 Transport hazard class(es)

ADR/RID: - IMDG: - IATA: -

14.4 Packaging group

ADR/RID: - IMDG: - IATA: -

14.5 Environmental hazards

ADR/RID: no IMDG Marine pollutant: no IATA: no

14.6 Special precautions for user

Land transport: Not regulated

Sea transport: Not regulated

Air transport: Not regulated

Inland waterway transport: Not regulated

Rail transport: Not regulated

14.7. Maritime transport in bulk according to IMO instruments

Not applicable

SECTION 15: REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU Regulations:

Authorisations and/or restrictions on use:

Authorisations: Not regulated

Restrictions on use: Not regulated

Other EU regulations:

Information according 1999/13/EC about limitation of emissions of volatile organic compounds (VOC-guideline): Not regulated

National regulations:

Listing	Ellagic acid
United States Toxic Substances Control Act (TSCA) Inventory	Not Listed
New Zealand Inventory of Chemicals (NZIoC)	Not Listed
Philippines Inventory of Chemicals and Chemical Substances (PICCS)	Not Listed
Australian Inventory of Industrial Chemicals (AIIC)	Not Listed
Canada Domestic Substances List (DSL)	Not Listed
Canada Non-Domestic Substances List (NDSL)	Not Listed
Vietnam National Chemical Inventory (NCI)	Not Listed
Chinese Chemical Inventory of Existing Chemical Substances (China IECSC)	Not Listed
China Catalog of Hazardous chemicals 2015	Not Listed
Korea Existing Chemicals List (KECL)	Not Listed

15.2 Chemical Safety Assessment

No Chemical Safety Assessment has been carried out for this substance/mixture by the supplier

SECTION 16: OTHER INFORMATION

Version information:

Version 2.0: Updated in accordance with Regulation (EC) No.1907/2006, (EC) No.1272/2008 and (EC) No. 2015/830. Add available data for SECTION 9, 11, 12 and 15.

Abbreviations and acronyms:

LD, lethal dose

EC, effect concentration

NOAEL, no-observed-adverse-effect level

NOEC, no-observed-effect-concentration-concentration

NOEL, No Observed Effect Level

PBT, persistent, bioaccumulative, and toxic

vPvB, very persistent, very bioaccumulative

DNELs, Derived No Effect Levels

PNECs, Predicted No Effect Concentrations

ADR: European Agreement concerning the International Carriage of Dangerous Goods by Road

RID: Regulation concerning the International Carriage of Dangerous Goods by Rail

IMDG: International Maritime Dangerous Goods

IATA: International Air Transportation Association

Key literature references and sources for data

ECHA

<https://echa.europa.eu/registration-dossier/-/registered-dossier/9662>

Disclaimer

The above information is believed to be correct but does not purport to be all inclusive and shall be used only as a guide. The information in this document is based on the present state of our knowledge and is applicable to the product with regard to appropriate safety precautions. It does not represent any guarantee of the properties of the product. SPEC-CHEM IND shall not be held liable for any damage resulting from handling or from contact with the above product. See reverse side of invoice or packing slip for additional terms and conditions of sale.

End of SDS